Space Race Analysis
Brief

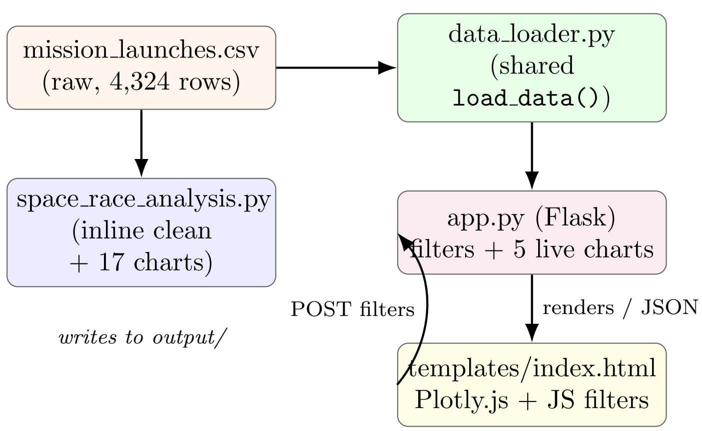
Generated explainer for Salman Adnan's portfolio

1 What it does

Analyses 4,324 space mission launches from 1957 to 2020, scraped from nextspaceflight.com, to answer: who launches the most, what does a launch cost, has launching gotten safer over time, and how did the USA and USSR compare during the Cold War. The project has two parts that read the same cleaned data:

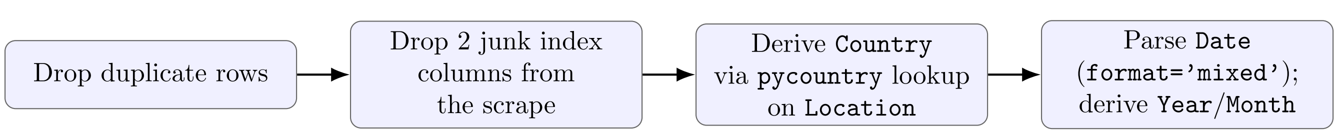
1. `space_race_analysis.py` — a static script producing 17 chart files (15 PNG, 2 interactive Plotly HTML) in `output/`, plus console summary tables.
2. `app.py` + `templates/index.html` — a Flask web dashboard with live organisation/country/year filters, re-slicing the real data on every request.

2 Architecture



Both `space_race_analysis.py` and `data_loader.py` independently apply the same cleaning steps (deliberate duplication, not a bug: the static script is left untouched so its console narration and behaviour never change), which is why the dashboard's numbers always match the static charts.

3 Data cleaning pipeline



The two junk columns exist because the CSV's header has a blank first field plus a field literally named `Unnamed: 0`; pandas auto-names the blank one `Unnamed: 0` too and resolves the name collision by calling the second one `Unnamed: 0.1`. The country lookup can't resolve historical entities like the former USSR (no longer in the current ISO country registry), so those rows get `Country = None` and are dropped from country-level charts only (choropleth, sunburst) — an intentional, documented undercount, not a bug.

4 The 17 static charts

#	Chart	What it shows
1–3	Bar charts	Launches per organisation; rocket status (active/retired); mission status breakdown.
4	Histogram	Launch price distribution ($0–1000M, missing prices dropped).
5	Choropleth (interactive)	Launches by country.
6	Sunburst (interactive)	Country → Organisation → Mission Status.
7–8	Bar charts	Launches per year; launches per month.
9	Line	Average price per year.
10–11	Bar / multi-line	Top 10 organisations; their launch cadence over time.
12–13	Multi-line / pie	NASA vs RVSN USSR over time and by share (Cold War).
14–15	Bar / line	Mission failures per year; failure rate (%) over time.
16	Bar	Launches per year, chronologically sorted.
17	Multi-line	Every organisation's launches per year.

A console-only table (not charted) shows total and average spend per organisation, for the 25 organisations with a nonzero reported price.

5 The Flask dashboard

`app.py` loads the cleaned data once at startup into an in-memory DataFrame (`DF`) and re-slices it on every request; nothing is precomputed per filter combination. Two routes:

- `GET /` — renders `index.html` with the full unfiltered dataset: organisation/country lists for the filter boxes, initial figures, and stats, all embedded as JSON in the page.
- `POST /api/filter` — takes {`organisations, countries, year_start, year_end`} as JSON, filters `DF` accordingly, and returns five fresh Plotly figures (serialised with `plotly.utils.PlotlyJSONEncoder` plus recomputed stats (total launches, distinct organisations, success rate, top organisation, year range).

The five live charts are: organisation bar (top 15), year line, mission-status bar (fixed colour per status), a world choropleth, and a country/organisation/status sunburst — each a filtered rebuild of its static-script counterpart. The page's JavaScript reads the filter controls, `fetch`es `/api/filter`, and calls `Plotly.react` to update the existing charts without a page reload. Three preset buttons exist: Reset (show everything), "USA vs USSR (1957–1991)" (NASA + RVSN USSR), and "Top 10 organisations" (computed from the real counts, not hardcoded).

6 Running it

```
python3 -m venv venv && source venv/bin/activate
pip install -r requirements.txt
python space_race_analysis.py   # writes 17 files to output/
python app.py                   # dashboard at http://127.0.0.1:5000/
```

No secrets or environment variables are required; `SHOW_PLOTS=1` is an optional toggle to also pop up each static chart interactively.

7 Notable limitations

- Country-level charts undercount historical USSR-era launches (lookup limitation, documented, not silent).
- Only ≈22% of launches report a price; price-based views drop missing rows rather than treating them as zero.
- The dashboard has no price filter/chart, unlike the static script.
- The static script is one long `main()`; no per-chart regeneration or isolated unit tests of the cleaning step.

8 Glossary

DataFrame pandas' in-memory table data structure with named columns.

CSV Comma-Separated Values, a plain-text spreadsheet format.

Flask A lightweight Python web framework; handles HTTP requests/routes.

Route A URL path a web server responds to (e.g. `/api/filter`).

JSON A text format for structured data, shared between the Flask server and the browser's JavaScript here.

ISO alpha-3 code A standardized three-letter country code (e.g. `USA`).

Choropleth A map shaded per region by a numeric value.

Sunburst chart A circular chart showing nested (hierarchical) categories as concentric rings.

NaN pandas' placeholder for a missing value.

Jinja2 — Flask's templating engine; fills {{ ... }} placeholders in HTML with real server-side values.

`Plotly.react` — Plotly.js's call to update an existing chart with new data in place, instead of rebuilding it from scratch.